

Multiple Choice

1. **Answer:** D

Options: A) ExoMars; B) Mars Exploration Rovers; C) Mars Science Laboratory; D) Phoenix Mars Lander

Note: The Phoenix Mars Lander is specifically designed to explore the Martian polar region, where it aims to excavate and analyze soil samples for the presence of water ice, making it the correct answer for the mission landing on May 25, 2008.

2. **Answer:** C

Options: A) red dwarfs.; B) pulsars.; C) black holes.; D) galaxies.

Note: The Schwarzschild radius defines the radius of the event horizon of a black hole, beyond which nothing can escape its gravitational pull, making it a fundamental concept in the study of black hole properties.

3. **Answer:** B

Options: A) nuclear fusion in the core; B) by contracting changing gravitational potential energy into thermal energy; C) internal friction due to its high rotation rate; D) chemical processes

Note: Astronomers believe that Jupiter generates its internal heat primarily through the process of gravitational contraction, where the planet's immense mass leads to a decrease in volume, converting gravitational potential energy into thermal energy as it collapses under its own gravity.

4. **Answer:** B

Options: A) Planets orbit around the Sun in nearly circular orbits in a flattened disk.; B) the equal number of terrestrial and jovian planets; C) the craters on the Moon; D) asteroids Kuiper-belt comets and the Oort cloud

Note: The nebular theory describes the process of solar system formation through the accretion of material from a protoplanetary disk, which does not inherently dictate a specific ratio of terrestrial to jovian planets, as this distribution is influenced by various factors such as distance from the Sun and the local conditions in the disk.

5. **Answer:** D

Options: A) Will the Sun become a black hole?; B) Is the universe infinitely large?; C) How old is the visible universe?; D) Are we alone in the universe?

Note: The Drake equation estimates the number of active, communicative extraterrestrial civilizations in the Milky Way galaxy, directly addressing the fundamental question of whether humanity is the only intelligent life in the universe.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: According to Stefan-Boltzmann Law, the power emitted by a blackbody is proportional to the fourth power of its temperature, so cooling the blackbody to half its original temperature results in a power emission decrease to $1/16$, while Wien's Displacement Law states that the peak emission wavelength is inversely proportional to temperature, thus doubling when the temperature is halved.

7. **Answer:** True

Options: A) True; B) False

Note: The statement is true because the relationship between actual brightness (luminosity) and apparent brightness is governed by the inverse square law, which allows astronomers to calculate the distance to an object using the formula that relates these two quantities.

8. **Answer:** True

Options: A) True; B) False

Note: Planet X is tidally locked, meaning its rotation period matches its orbital period around its sun, leading to one side perpetually facing the sun and the other side in constant darkness, thus eliminating the concept of a solar day.

9. **Answer:** True

Options: A) True; B) False

Note: Bolometric luminosity encompasses the entire spectrum of electromagnetic radiation emitted by an astronomical object, reflecting its total energy output across all wavelengths, thus affirming the statement as true.

10. **Answer:** False

Options: A) True; B) False

Note: Asteroids do not emit their own radiation; instead, they reflect sunlight, which is why they are visible in sky surveys.

Long Form Response

11. **Answer:** The leading hypotheses regarding the origin of the Moon include the Giant Impact Hypothesis, the Double Planet Hypothesis, and the Fission Theory, with the Giant Impact Hypothesis being the most widely accepted. This hypothesis posits that the Moon formed from the debris generated by a massive collision between the early Earth and a Mars-sized body. In contrast, the theory of the Moon splitting from the Earth due to tidal forces has not gained acceptance primarily because it fails to account for the angular momentum and compositional differences between the Earth and Moon. The fission theory suggests that the Moon was once part of the Earth and separated due to rapid rotation, but evidence indicates that such a scenario is unlikely given the current understanding of planetary formation and dynamics.

Note: correct because it identifies the Giant Impact Hypothesis as the most widely accepted explanation for the Moon's origin, while criticizing the Fission Theory for its inability to adequately explain the Moon's characteristics and formation process, particularly in relation to the physical dynamics of tidal forces.

12. **Answer:** Betelgeuse, a prominent red supergiant star located in the Orion constellation, holds significant importance in both cultural and scientific contexts. Physically, it is one of the largest and most luminous stars known, with a diameter approximately 1,000 times that of the Sun and a distinct reddish hue due to its low surface temperature of around 3,200 Kelvin. Culturally, Betelgeuse has been an object of fascination across various civilizations, often associated with myths and navigation due to its brightness and position in Orion, which itself symbolizes hunters in several cultures. In astronomical studies, Betelgeuse serves as a key subject for understanding stellar evolution, particularly the life cycles of massive stars and the dynamics of supernovae, as it is expected to end its life in a spectacular explosion that will offer insights into cosmic processes. Its impending transformation and variability also make it a vital reference point for astrophysical research, embodying the interplay between observational astronomy and theoretical models.

Note: correct because it effectively addresses the significance of Betelgeuse by detailing its physical characteristics as a massive red supergiant, its cultural relevance in mythology and navigation, and its contributions to astronomical studies, thus providing a comprehensive overview of the star's multifaceted importance within the Orion constellation.